

LOADZONE · REQUIREMENTS

LoadZone requirements

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DOCUMENT CONTROL

Document ID	LDZ-REQ-001	Version	0.4
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/loadzone		
Source	docs/03-requirements.md @ 16652fe		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from LDZ-CAL-001; R6 load basis set; R13 redefined to the two-puck kit (LDZ-DDR-001 D6)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

LoadZone requirements

These are the requirements for the concept, checked by calculation at TRL 3 in LDZ-CAL-001 v0.2. On 2026-09-25 Amish accepted the recommendations (LDZ-DDR-002), so this version restates three targets: R4 is now a gateway within 300 m of the bay rather than 1 km; R6 is restated with the finding that the wheels of a vehicle parked inside the bay lines straddle the puck, and the bond is two-part road-marker epoxy; and R11 is a daylight-only target. R13 covers the two-puck bay kit only, with the sign option costed separately (LDZ-DDR-001 D6, now decided). R12 uses FieldNode's published 100 mW allowance.

Table 1. Requirements

ID	Requirement	Target	Status at TRL 3 (LDZ-CAL-001)	Verification
R1	Detect the state of each vehicle slot	97 % or more of slot states correct over a day, for cars, vans and box trucks, with traffic passing in the next lane	At risk , not verifiable at TRL 3: a line-dipole model puts a van at 10.0 μ T, a weak-steel box truck at 3.85 μ T and next-lane traffic at 0.46 μ T against a 3 μ T threshold, on an assumed calibration	Field trial against a manual count or time-lapse survey (TRL 5)
R2	Report changes quickly	Slot state at the server within 60 s of a vehicle arriving or leaving	Met on paper at SF10 or faster: 17.7 s typical, 38.6 s worst at SF9, 59.2 s at SF10	Timing calculation (TRL 3)
R3	Long battery life	5 years or more at 100 state changes a day plus hourly heartbeats	Met on paper: 7.7 years at SF9 on two AA-size Li-SOCl ₂ cells derated 40 %; 2.8 years at SF12	Power budget calculation (TRL 3)
R4	Reach a gateway from road level	Uplinks received by a gateway within 300 m of the bay in a street canyon, with a van parked over the puck (restated from 1 km, LDZ-DDR-002)	Met on paper: 353 m out of sight at SF9 with a 10 dB fade margin, 1.18 times the distance	Link budget (TRL 3), then field survey
R5	Stay within network fair use	30 s or less of uplink airtime per puck per day (TTN fair use)	At risk : 25.5 s a day at SF9, but 51.0 s at SF10 and 183.9 s at SF12	Airtime calculation (TRL 3)
R6	Survive traffic	Withstand a maneuvering wheel crossing the puck at 49 kN (half of a 10 t single axle), 64 kN with a 1.3 dynamic factor, and repeated drive-overs by loaded trucks, without cracking or debonding, on a two-part road-marker epoxy bed. The puck sits at the slot center so that the wheels of a vehicle parked inside the bay lines straddle it (restated, LDZ-DDR-002)	At risk : crushing met on paper (factor 5.4 on the cast dome, 2.0 on the potting); parked wheels clear the puck by at least 275 mm; bond shear under braking 1.51 MPa against about 1.0 MPa assumed for epoxy on asphalt	Load calculation (TRL 3), then load and pull-off test (TRL 4, on hold)
R7	Safe, low profile	Height 35 mm or less above the road, rounded edges, high-visibility color	Met on paper: 31 mm high, 150 mm diameter, 6 mm crown radius, yellow dome	Parametric model and drawing LDZ-DWG-001
R8	Weather and temperature	Sealed to IP68; operate from -25 to +70 °C road surface temperature; resist salt, oil and fuel	Met by design: parts rated -40 to +85 °C with an 85 °C pulse capacitor; fully potted. Sealing not verifiable at TRL 3	Datasheet review (TRL 3), then immersion test
R9	Privacy by design	Only slot state, timestamps and device health leave the device; no images, audio or vehicle identifiers	Met by design: a magnetometer cannot read plates or faces	Design review
R10	Open data	Bay state available through an open API that maps onto the Curb Data Specification Events and Metrics APIs (Open Mobility Foundation)	Met by design: events map onto <code>park_start</code> , <code>park_end</code> , <code>scheduled_report</code> , <code>comms_lost</code> and <code>comms_restored</code>	Design review (TRL 3)

ID	Requirement	Target	Status at TRL 3 (LDZ-CAL-001)	Verification
R11	Sign legible to drivers (option)	Free-slot count legible at 25 m by day (daylight only, restated, LDZ-DDR-002); at night drivers use the data feed	Met on paper: about 28 m by day at full contrast with 78 mm digits	Legibility calculation (TRL 3)
R12	Sign power (option)	Sign runs within the host FieldNode's sensor energy allowance	Met on paper: 0.42 Wh a day (17.7 mW) against FieldNode's published 100 mW; needs a private network for its 200 downlinks a day (LDZ-DDR-002, O5)	Power budget (TRL 3)
R13	Low cost	Two-puck bay kit \$120 or less in parts at quantity 1; the sign option and its host FieldNode are costed separately and excluded	Met on paper: \$114.00. Sign option \$97.00 plus a FieldNode at \$126.00	Priced BOM
R14	Quick install without road works	One puck bonded in 15 min or less by a two-person crew; removable without damaging the road	Not verifiable at TRL 3	Install trial (TRL 4 or later)
R15	Tamper and theft resistant	No external screws; removal needs a heat gun or scraper; device reports removal	Met by design: bonded, no fasteners; removal flagged by a sudden field and orientation change	Design review (TRL 3)

■ Requirements at risk at TRL 3

No requirement is now not met. Before LDZ-DDR-002, R4 (1 km) and R11 (night) were not met; both are restated and met on paper.

- **R1 (detection)** is at risk for high-clearance trucks and rests on an assumed signal; only field data can settle it.
- **R5 (airtime)** is not met at SF10 or slower; R2 and R3 also fail at SF11 and SF12, so pucks must be placed within SF9 reach of a gateway, which the 300 m gateway distance in R4 supports.
- **R6 (traffic)** is at risk on debonding under braking, even with the epoxy bed; a pull-off and shear test on asphalt would settle it, but that is TRL 4 work and on hold.

■ Assumptions

- A vehicle slot is about 7 m long and the puck sits near its center, 1.3 m from the curb face (estimates; bay layouts vary by city).
- Loading bay turnover of up to 50 vehicles a slot a day, or 100 state changes, based on most delivery stops lasting 30 minutes or less ([Urban Freight Lab, 2019](#)).
- Each 12-byte LoRaWAN uplink costs about 14.2 mA·s at SF9 with FieldNode's radio currents (LDZ-CAL-001). If the band is US915, the payload is packed to 11 bytes (LDZ-DDR-002); the band is still open.
- The wheel load is half of a 10 t single axle (49 kN); a vehicle parked inside the bay lines straddles the puck, so wheel loads come from maneuvering (LDZ-CAL-001, section 7).